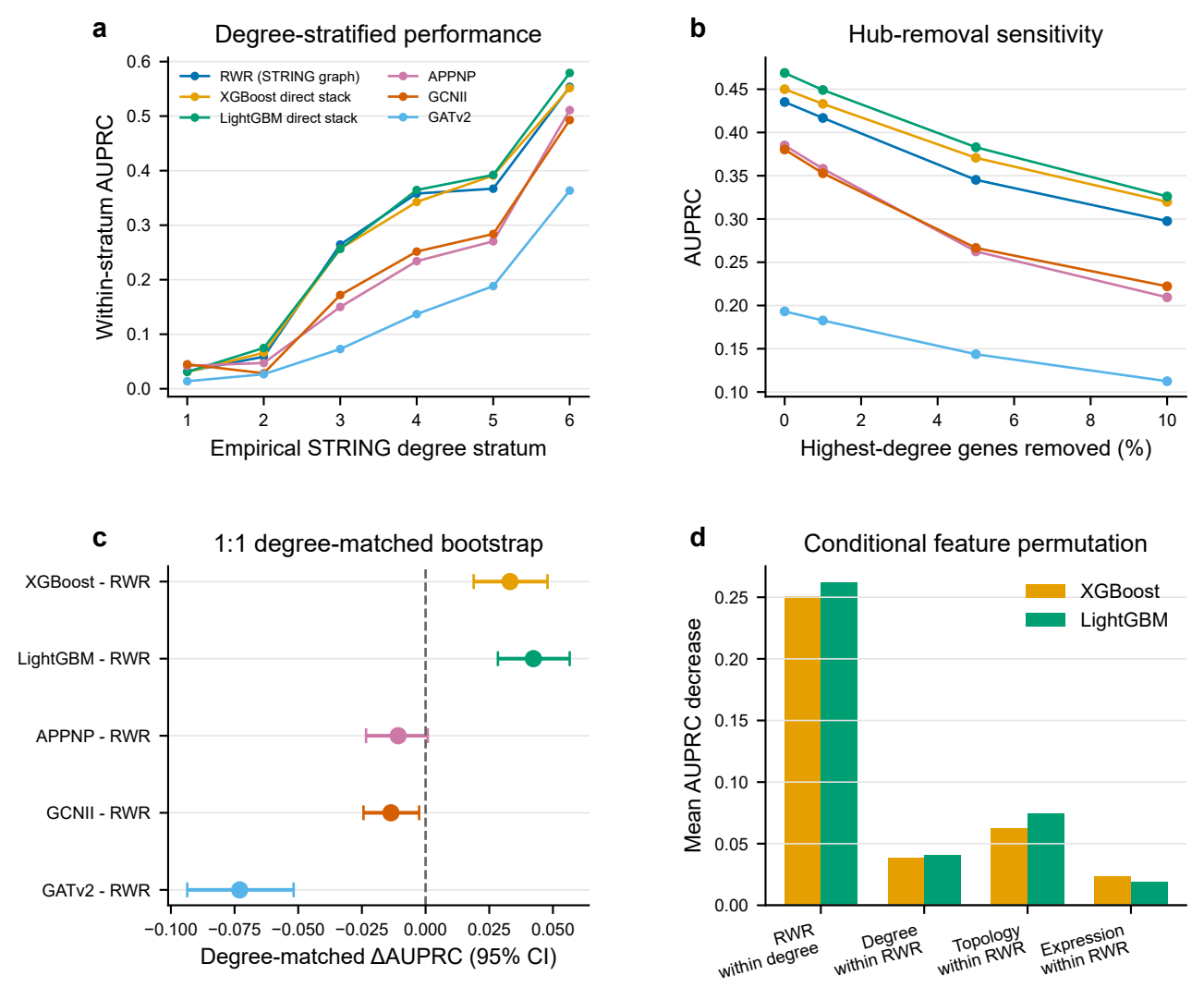
a
Degree-stratified performance
RWR (STRING graph)
XGBoost direct stack
LightGBM direct stack
APPNP
GCNII
GATv2
Within-stratum AUPRC
Empirical STRING degree stratum
b
Hub-removal sensitivity
AUPRC
Highest-degree genes removed (%)
c
1:1 degree-matched bootstrap
XGBoost - RWR
LightGBM - RWR
APPNP - RWR
GCNII - RWR
GATv2 - RWR
Degree-matched ΔAUPRC (95% CI)
d
Conditional feature permutation
XGBoost
LightGBM
Mean AUPRC decrease
RWR within degree
Degree within RWR
Topology within RWR
Expression within RWR